

Transistor-Level CMOS Design and Optimization of a 4-Bit Array Multiplier using Tanner EDA

Final Report

(f) Optimize the array multiplier design such that you are able to reduce the critical path delay, or the area, or the power consumption. Demonstrate the functionality through waveforms for 2-3 different input combinations. Estimate the critical path delay through simulations. Calculate the power consumption and the transistor area for this implementation.

Ans:

Initially, the XOR gate is optimized to minimize the number of transistors, achieving an efficient design. The optimized design is shown below.

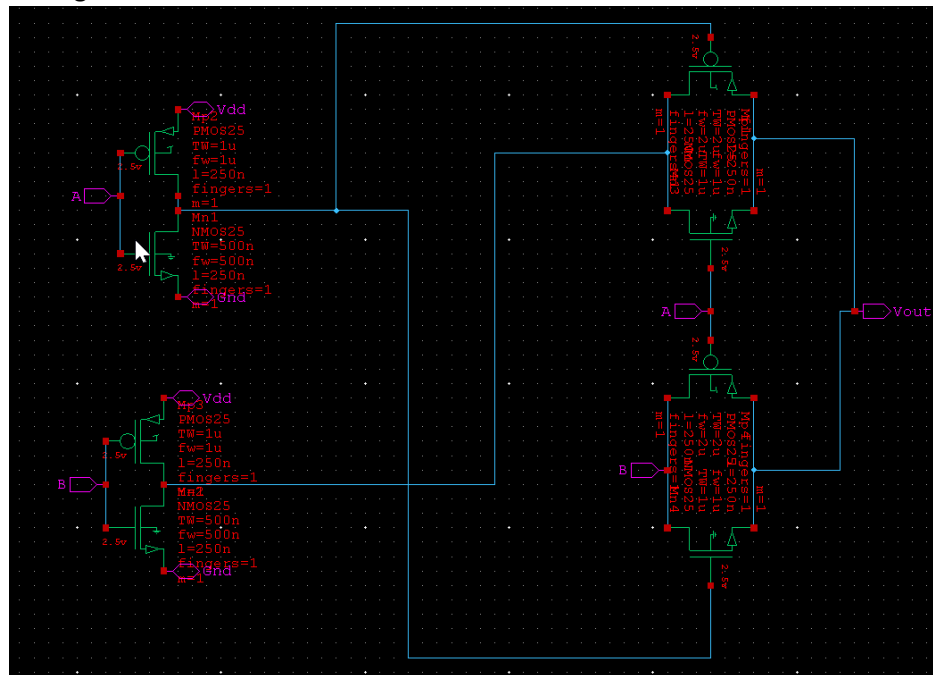


Fig-1: Optimized XOR gate

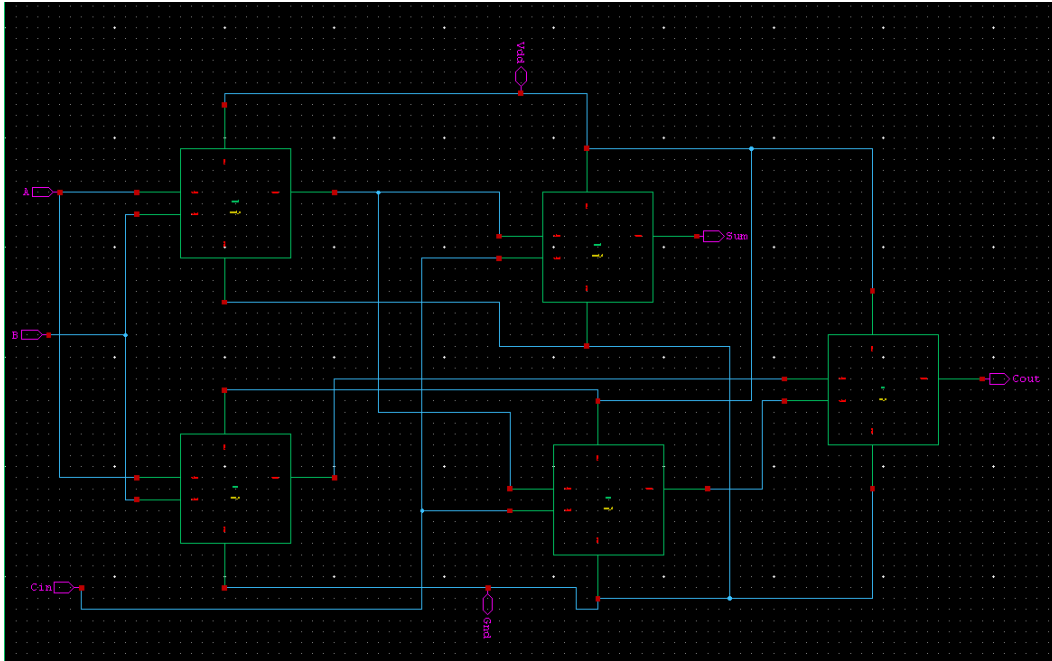


Fig-2: Full adder with optimized XOR gate

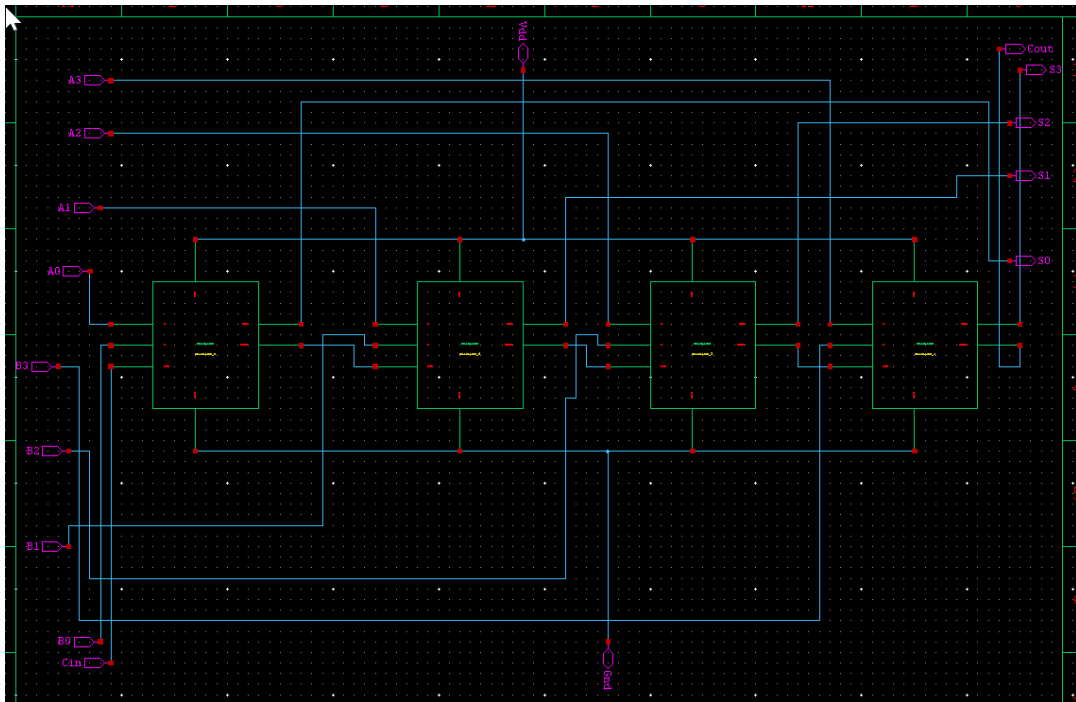


Fig-3: Four-Bit adder with optimized XOR gate

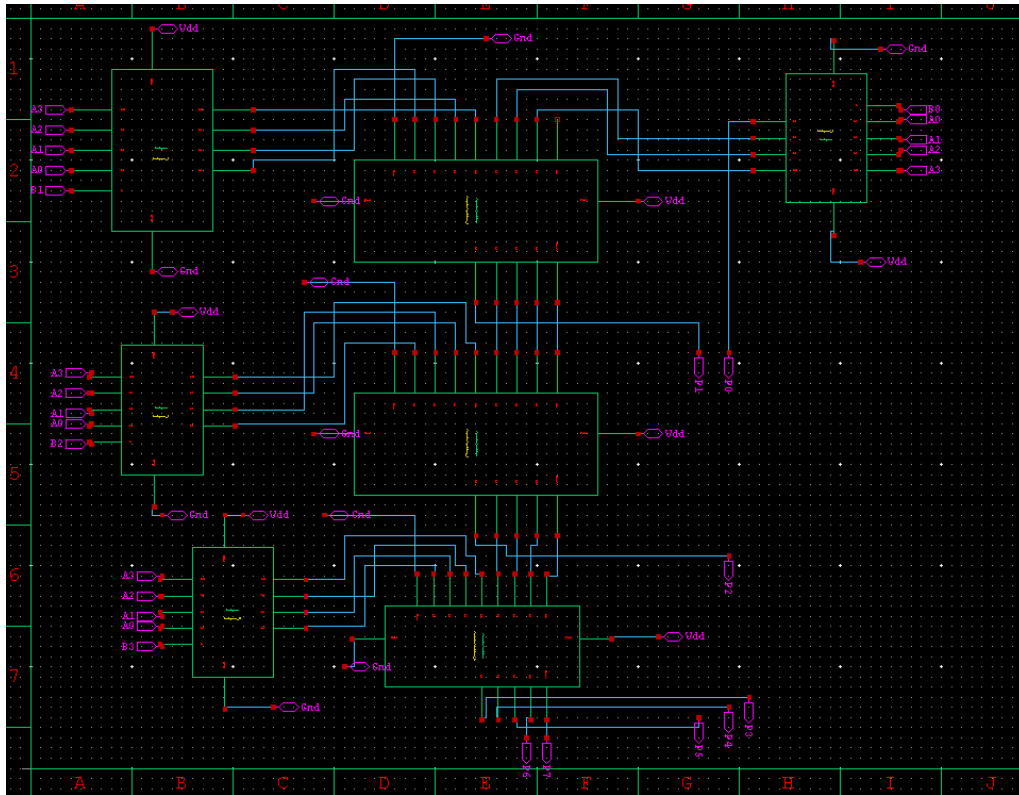


Fig-4: Four-Bit array multiplier with optimized XOR gate

In this design, the number of transistors used in the XOR gate was optimized, which significantly reduced the overall transistor count in the arithmetic circuits. As a result, the full adder implementation requires only 38 transistors. Extending this design, a 4-bit ripple carry adder utilizes a total of 152 transistors, while a 4-bit array multiplier requires 552 transistors.

The reduction in transistor count directly impacts several key performance parameters. Firstly, a lower number of transistors leads to a decrease in silicon area, making the design more compact and cost-effective for VLSI implementation. Secondly, power consumption is reduced due to fewer switching elements and lower parasitic capacitances, which minimizes both dynamic and static power dissipation. Additionally, the critical path delay is improved because fewer logic stages and reduced capacitance result in faster signal propagation.

Overall, optimizing the XOR gate and minimizing transistor usage enhances the efficiency of the entire system, achieving a better trade-off between area, power, and speed. This makes the design more suitable for high-performance and low-power digital applications.

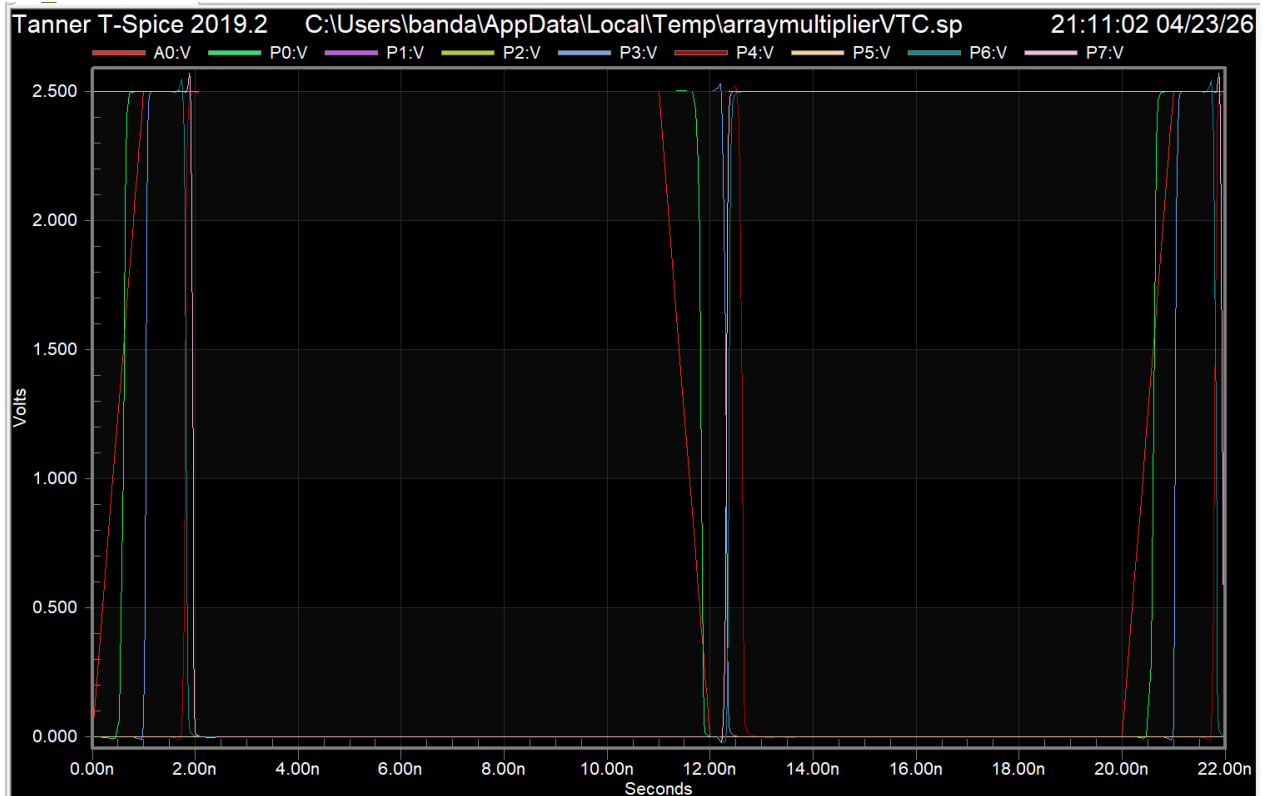


Fig-5: Output waveforms of the four-bit array multiplier

In above waveform figure the inputs are taken

A = 0011 $\rightarrow$ 1011

B = 1110 $\rightarrow$ 1101

The output are

P = 00101010 $\rightarrow$ 10001111

Propagation delay $t_p = 5.6825\text{ns}$ which is less that obtained in non-optimized case

Power Consumption = -929.1682uW

```
Vdd Vdd Gnd DC 2.5

* ----- Inputs -----
* A inputs
Va0 A0 Gnd PULSE(0 2.5 0ns 1ns 1ns 10ns 20ns)
Va1 A1 Gnd DC 0
Va2 A2 Gnd DC 2.5
Va3 A3 Gnd DC 2.5

* B inputs
Vb0 B0 Gnd DC 2.5
Vb1 B1 Gnd DC 2.5
Vb2 B2 Gnd PULSE(2.5 0 0ns 1ns 1ns 10ns 20ns)
Vb3 B3 Gnd DC 2.5

* ----- Analysis -----
.tran 0.1ns 22ns

* ----- Output -----
.plot tran V(A0) V(P0) V(P1) V(P2) V(P3) V(P4) V(P5) V(P6) V(P7)
.measure tran avg_power AVG (V(Vdd)*I(Vdd)) FROM = 1n TO = 20n
.measure tran max_power MAX (V(Vdd)*I(Vdd)) FROM = 5n TO = 20n
.measure tplt trig V(A0) val = 1.25 rise = 1 targ V(P7) val = 1.25 rise=1
.measure tplt trig V(A0) val = 1.25 fall = 1 targ V(P7) val = 1.25 fall=1
.end
```